



Spark Team

أسئلة سنوات سابقة لمادة

نظرية التحكم

شكوت إلى وكيع سوء حفظي قأرشدني إلى ترك المعاصي
وأخبرني بأن العلم نور ونور الله لا يؤتّى لعاصي



AL-BALQA' APPLIED UNIVERSITY

Faculty of Engineering Technology

Department of Mechatronics

Second semester , first, 2010 , 20 points



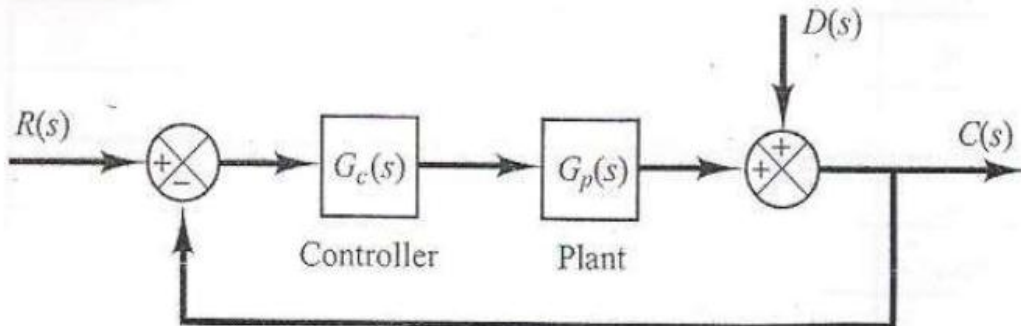
Instructor name

Dr Ibrahim Al-abbas

Dr Ma'moun Janajrah

اسم الطالب ..

Q1: Obtain the expression for the output $C(s)$ when both the reference input and disturbance input are present.



Q2: Consider the system described by

$$\ddot{y} + 3\dot{y} + 2y = u$$

$$s^2 Y(s) + 3sY(s) + 2Y(s) = U(s)$$

Derive the state space of the system. Where $y(t)$ is the output.

Q3: Consider the system described by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -4 & -1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = [1 \ 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Obtain the transfer function of the system.

See page (2)

$$\frac{s^2 + s - 0.25}{(s - 0.5)(s - 0.5)} = \frac{s^2 + s - 0.25}{s^2 - s - 0.25}$$

أستغفر الله العظيم

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Department of Mechatronics

Spring Semester, Second 2010, 20 points



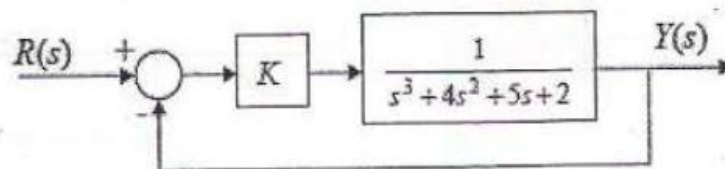
Instructor name

Dr Ibrahim Al-abbas ☐

Dr Ma'moun Janajrah ☒

اسم الطالب ..

Q1. Consider the proportional feedback control system shown in the Figure.



1. Determine the value of K to reduce the steady-state error to 4% of a constant input. $e_{ss} = 0.04$
2. Use Routh-Hurwitz criterion to determine the value of K over which the system is stable.

Q2. Obtain the unit impulse response and the unit step response of a unity feedback system whose open-loop transfer function is

$$G(s) = \frac{2s + 1}{s^2}$$

Q3 Consider the following figures that show the effect of P, PI, PD and PID controllers on the system performance. Study such effects and complete the following table. Use the suitable word in each table cell, such as increase, decrease, eliminate, small change, etc.

	Rise time	Maximum overshoot	Settling time	Steady state error
P	Decrease	increase	small change	Decrease
PI	Decrease	increase	increase	Eliminate
PD	Decrease	increase		
PID	small change	Decrease	Decrease	small change

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Second semester , first, 2010 , 20 points



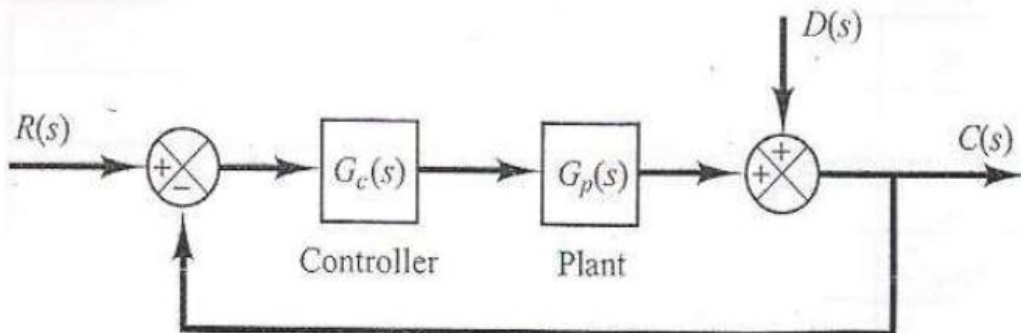
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Derive the state space of the system. Where $y(t)$ is the output.

Q3: Consider the system described by

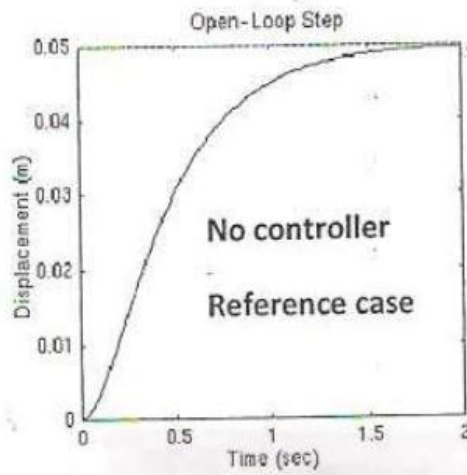
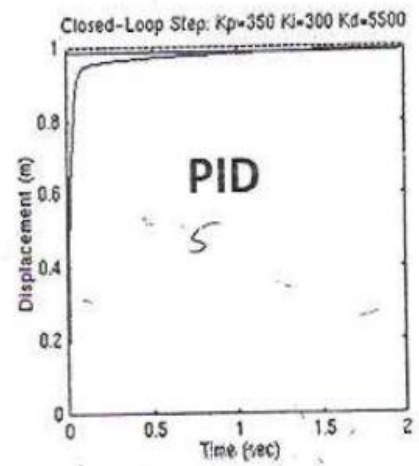
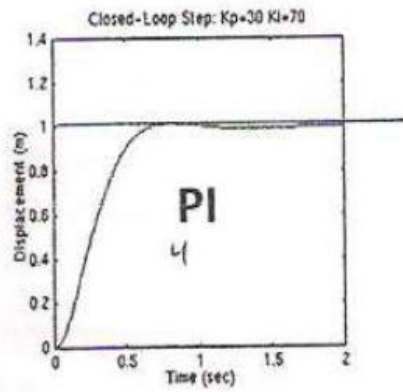
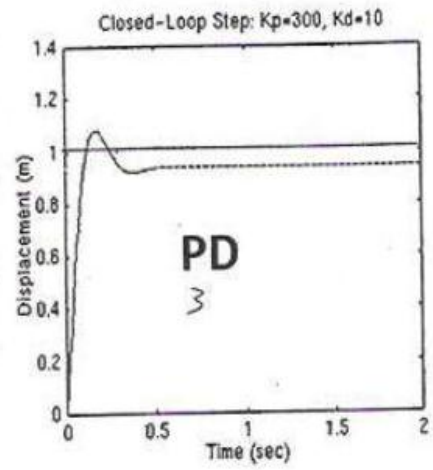
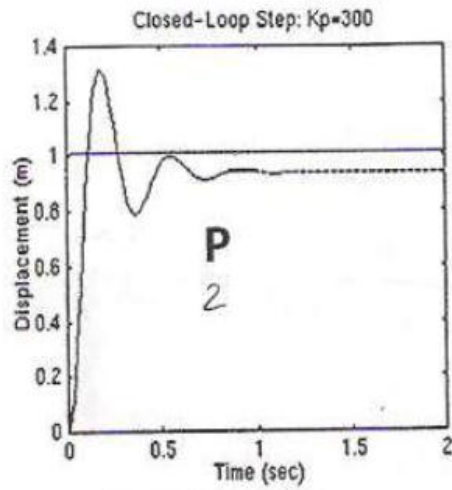
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -4 & -1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Obtain the transfer function of the system.

See page (2)

$$\frac{s^2 + s - 0.25}{(s - 0.5)(s - 0.5)} = \frac{s^2 + s - 0.25}{s^2 - s - 0.25}$$



(1)

د. مأمون حناجر

طريقة تحكم

الاسم:

$$Q1) G(s) = \frac{Y(s)}{R(s)} = \frac{\frac{K}{s^3 + 4s^2 + 5s + 2}}{1 + \frac{K}{s^3 + 4s^2 + 5s + 2}}$$

1)

$$= \frac{K}{s^3 + 4s^2 + 5s + 2} + \frac{s^3 + 4s^2 + 5s + 2}{s^3 + 4s^2 + 5s + 2 + K}$$

$$G(s) = \frac{K}{s^3 + 4s^2 + 5s + 2 + K}$$

$$Q2) \frac{K}{s^3 + 4s^2 + 5s + 2 + K}$$

$$w_{ss} = G(0) = \frac{K}{2 + K} = 0.04$$

$$(2 + K)0.04 = K$$

$$0.08 + 0.04K = K$$

$$\therefore K = 0.083$$

2)

$$\begin{array}{l} s^3 : 1 \quad 5 \\ s^2 : 4 \quad 2+K \\ s^1 : \frac{18-K}{4} \quad 0 \\ s^0 : 2+K \end{array}$$

$$\frac{4 \times 5 - (2+K)}{4} = \frac{18-K}{4}$$

$\therefore$ to be the system stable should be

$$\frac{18-K}{4} > 0 \text{ and } K > 0$$

$$\therefore 0 < K < 18$$

Q2

$$G(s) = \frac{2s+1}{s^2}$$

* impulse response ($R(s) = 1$)

$$\frac{2s+1}{s^2} = \frac{A}{s} + \frac{B}{s^2}$$

$$2s+1 = As^2 + Bs \quad \text{--- (1)}$$

$$s = -1 \quad \therefore -2+1 = +A - B$$

$$1 = A - B \Rightarrow A = 1 + B$$

$$2s+1 = As^2 + Bs$$

$$s = 1$$

$$3 = A + B$$

$$3 = 1 + B + B$$

$$4 = 2B \Rightarrow \boxed{B = 2}$$

$$\therefore \boxed{A = 3}$$

$$\frac{2s+1}{s^2} = \frac{A}{s} + \frac{B}{s^2}$$

$$\mathcal{L}^{-1}\left\{\frac{2s+1}{s^2}\right\} = \frac{3}{s} + \frac{2}{s^2}$$

$$\mathcal{L}^{-1} = 3 + 2t$$

$$\therefore g(t) = 3 + 2t$$

impulse response

* unit step response

$$R(s) = \frac{1}{s}$$

$$\frac{1}{s} = \frac{A}{s} + \frac{B}{s^2}$$

$$\therefore G(s) = \frac{2s+1}{s^2}$$

next



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الرقم الجامعي

First Exam

اسم الطالب

Q1) (8 points) given $y'''(t) + 2y''(t) + 3y'(t) + 4y(t) = r(t) + u(t)$. Determine

$$1- Y(s)/R(s) =$$

$$2- Y(s)/U(s) =$$

3- The state space matrices

A=

B=

C=

D=

Q2) (6 points)

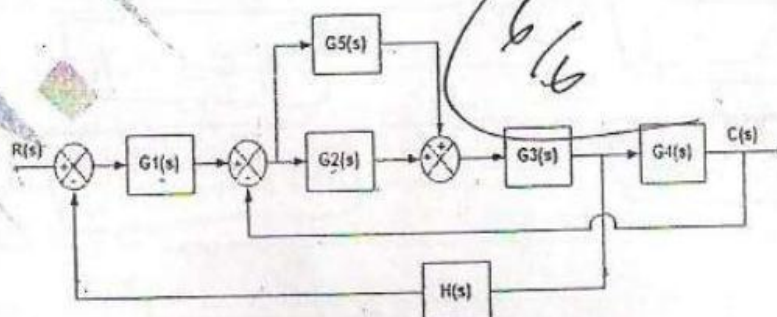
1- write the partial fraction expansion of

$$F(s) = \frac{s+2}{s^2(s+4)}$$

2- Inverse Laplace the given expression

$$C(s) = \frac{2}{s+5} + \frac{s+2}{(s+2)^2 + 3}$$

Q3) (6 points) reduce, in steps, the following block diagram



$$\text{1a)} \quad \mathcal{L} [\ddot{y}(t) + 2\dot{y}(t) + 3y(t) + 4y(t)] = \mathcal{L}[r(t) + u(t)]$$

$$3sY(s) + s^3Y(s) + 2s^2Y(s) + 4Y(s) = R(s) + U(s)$$

$$Y(s) [s^3 + 2s^2 + 4 + 3s] = R(s) + U(s) \rightarrow \text{Superposition}$$

$$\boxed{\frac{Y(s)}{R(s)} = \frac{1}{s^3 + 2s^2 + 4 + 3s}}$$

1c

2a)

$$Y(s) [s^3 + 2s^2 + 4 + 3s] = U(s)$$

$$\boxed{\frac{Y(s)}{U(s)} = \frac{1}{s^3 + 2s^2 + 4 + 3s}}$$

(3)

$$\dot{\bar{x}} = A\bar{x}(t) + Bgu(t)$$

$$y(t) = C\bar{x}(t) + Du(t)$$

$$\boxed{x_1 = y(t)}$$

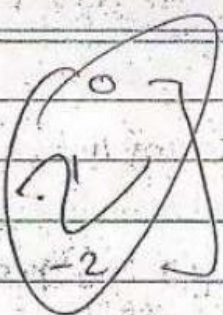
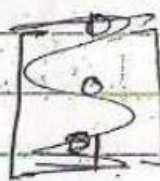
$$\boxed{x'_1 = x_2(t)}$$

$$\boxed{x_2 = \dot{y}(t)}$$

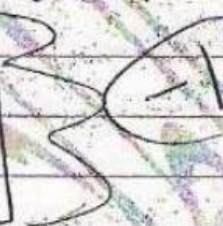
$$\boxed{x'_2 = x_3(t)}$$

$$x_3 = \ddot{y}(t)$$

$$\boxed{x'_3 = \ddot{\dot{y}}(t)}$$

A = $\begin{bmatrix} 0 & 1 \\ 0 & 0 \\ -4 & -3 \end{bmatrix}$  

B = $\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$ 

$y = Cx(t) + Du(t)$
 $C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$ 
 $D = \begin{bmatrix} 0 \end{bmatrix}$

Q2:- $\frac{1}{s^2(s+4)} = \frac{A}{s} + \frac{2B}{(s)^2} + \frac{C}{(s+4)}$

$(s+2) = A s(s+4) + B(s+4) + C s^2$

$2 = 4B$

$B = \frac{1}{2}$

$C = -1$

$$3 = \frac{5A + 5B + C}{s-1}$$

$$5A + 2.5 + \frac{-1}{8} = 3$$

$$5A = 3 - 2.5 + \frac{1}{8}$$

$$A = \frac{1}{8}$$

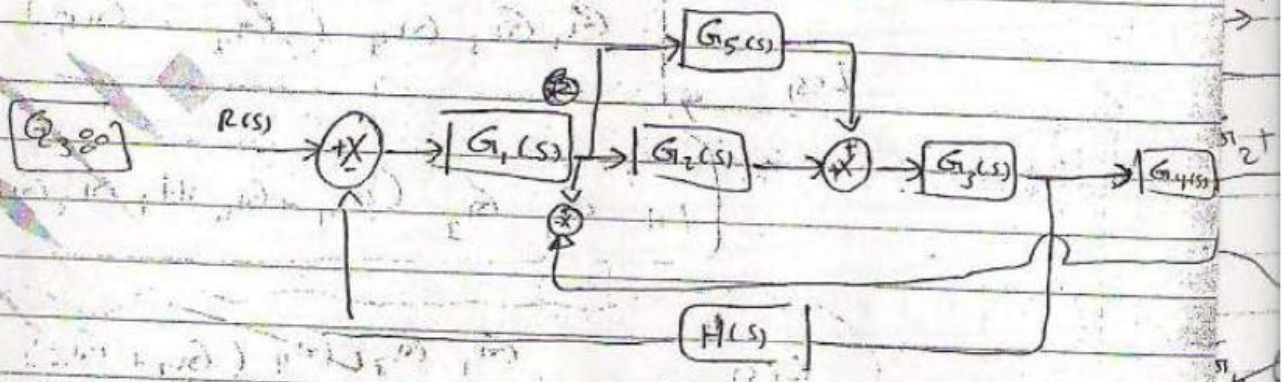
$$= \frac{\frac{1}{8}}{s} + \frac{\frac{1}{2}}{s+2} + \frac{\frac{-1}{8}}{s+4}$$

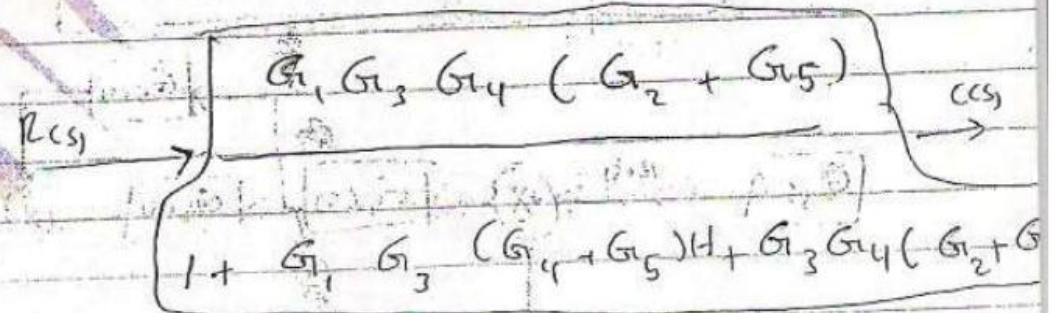
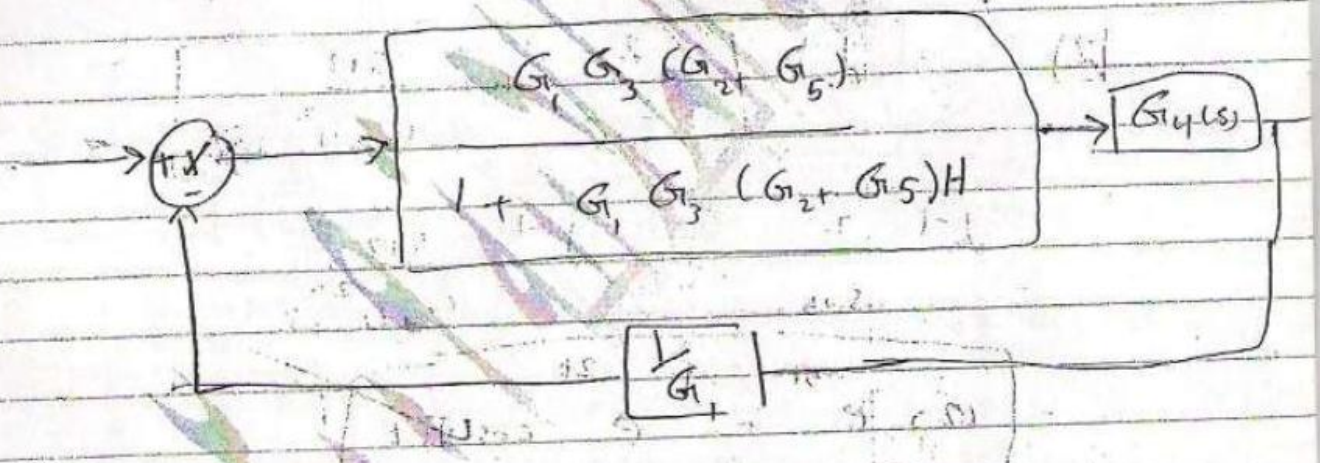
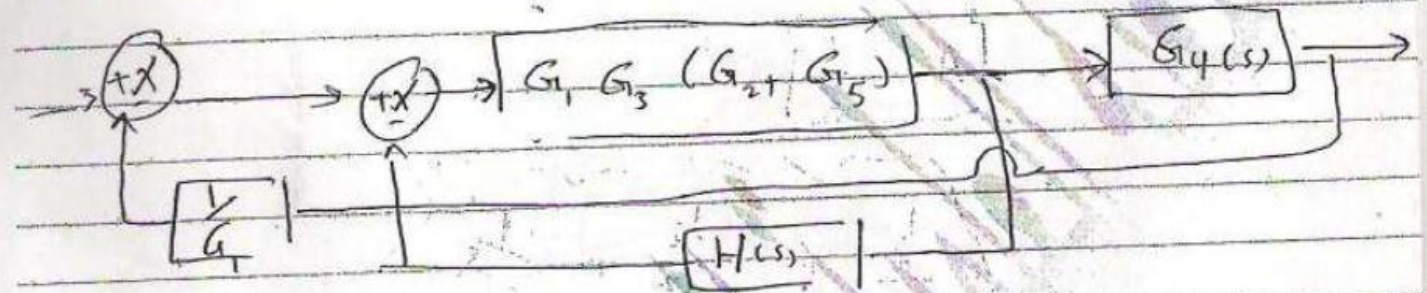
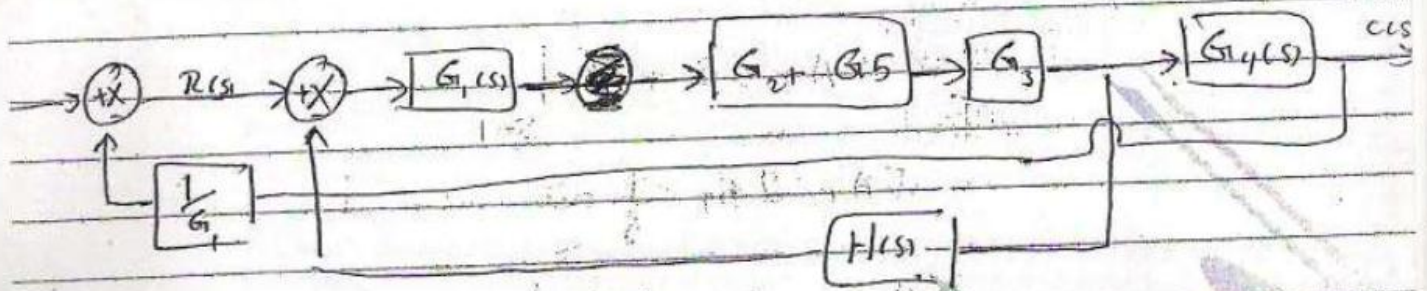
(2)

$$\mathcal{L}^{-1} \left[\frac{2}{s+5} + \frac{s+2}{(s+2)^2 + 3} \right]$$

$$\mathcal{L}^{-1} \frac{2}{s+5} + \mathcal{L}^{-1} \frac{s+2}{(s+2)^2 + 3}$$

$$(2) e^{-5t} + e^{-2t} \cos \sqrt{3} t$$





$$\frac{C(s)}{R(s)} = \frac{G_1 G_3 G_4 (G_2 + G_5)}{1 + G_1 G_3 (G_4 + G_5) H + G_3 G_4 (G_2 + G_5)}$$

خدمتكم عبادة نتقرب بها الى الله اخوانكم معاذ ابو صليح و محمد جوايرة